MIL-M-13685F (MU)
3 August 1973
SUPERSEDING
MIL-M-13685E (MU)
28 September 1966

MILITARY SPECIFICATION

MINE, AT, NM, M19 WITH ARMING WRENCH M22 LOADING ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of mine designated as Mine, Anti-tank, nonmetallic, M19.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

C8830877 - Box, Wirebound, Packing, Ammunition for Mine, AT, NM, M19 with Fuze, Mine, AT, NM, M606.
C8830878 - Support, Packing, Ammunition for Mine, AT, NM, M19 with Fuze, Mine, AT, NM, M606.
9209328 - Mine, AT, NM, M19.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.1.1 Scrap.-Scrap formed in kettles, pails, risers, and loading machines used in the pouring and cooling operations, may be remelted and reused; however, the scrap shall not be contaminated in any way and shall comply with all requirements for the grade of high explosive specified except for form. Scrap from floor sweepings, ventilators, and suction apparatus shall not be used.

3.2 Assembly.-The mines shall comply with all requirements specified on Drawing (dwg.) 9209328, and with all requirements specified in applicable specifications.

3.3 Functioning.-The mine shall function satisfactorily, as evidenced by formation of a crater with a minimum depth of one foot and a minimum diameter of five feet when tested as specified.

3.4 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dents, gouges, holes, porous areas, distortion, dirt, rust, chips, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

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4.2.1 Lot Formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of mine assemblies shall contain:

a. Parts of one interfix lot number from one manufacturer.
b. Activators of one interfix lot number from one manufacturer.
c. Fuzes of one interfix lot number from one manufacturer.
d. RDX or Composition A5 of one interfix lot number from one manufacturer.
e. Composition B4 of one interfix lot number from one manufacturer.
f. Loaded detonator holder assemblies of one interfix lot number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's

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to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Plastic Parts Assembly (see dwg. 9209330 covering a detail of dwg. 9209329.)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Assembly cracked or split.....	Visual	01001
102.	Carrying cord missing.....	Visual	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship in charge cavity or well (see 3.4)..	Visual	01003

4.2.2.2 Assembly, less booster well cover (see dwg. 9209329 covering a detail of dwg. 9209328).

Categories	Defects	Metnod of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Booster pellet missing.....	Visual	02001
Minor:	None defined		

4.2.2.3 Assembly, less fuze (see dwg. 9209329 covering a detail of dwg. 9209328).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:	AQL 0.40 percent		
101.	Booster well cover or filling hole cover missing, damaged or not cemented for 360 degrees..	Visual	03001
102.	Weight, minimum (min).....	Balance	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	03003

4.2.2.4 Assembly (see dwg. 9209328).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Assembly cracked or split.....	Visual	04001
102.	Fuze assembled with detonator holder loading assembly, in lieu of shipping plug.....	Visual	04002
103.	Fuze assembly missing or damaged.	Visual	04003
Minor:	AQL 1.50 percent		
201.	Carrying cord broken or frayed...	Visual	04004
202.	Marking misleading or unidentifiable.....	Visual	04005
203.	Evidence of poor workmanship (see 3.4).....	Visual	04006

4.2.2.5 Box, packing ammunition, prior to sealing (see dwg. 8830877).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	One or more detonator holder loading assemblies missing.....	Visual	05001
102.	Wrench missing or one or more mines missing.....	Visual	05002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	05003

4.2.2.6 Box, packing, ammunition, sealed (see dwg. 8830877).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Hardware missing, broken or loose.....	Visual-Manual	06001
102.	Strapping missing, broken or loose.....	Visual-Manual	06002
103.	Contents of box exposed.....	Visual	06003
104.	D.O.D. symbol misleading or unidentifiable.....	Visual	06004
105.	Board broken.....	Visual	06005
Minor:	AQL 2.50 percent		
201.	Car seal missing, unsealed or improperly positioned.....	Visual-Manual	06006
202.	Handle missing or insecure.....	Visual-Manual	06007
203.	Hardware improperly engaged.....	Visual-Manual	06008
204.	Contents loose (when box is tilted).....	Manual	06009
205.	Strapping improperly assembled...	Visual	06010
206.	Marking misleading or unidentifiable.....	Visual	06011

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4.2.3 Testing

4.2.3.1 Functioning (see 3.3) Major Defect - Code No. 07001.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter J, double sampling plan, and an AQL of 1.5 percent. The samples drawn shall have the M606, AT, NM, Mine fuze assembled to them. The test shall be performed as specified in 4.3.1 using equipment in accordance with 4.2.4. Any assembly which fails to comply with the specified requirement shall be classed defective. Failure caused by malfunctioning of the fuze will be disregarded.

4.2.3.1.1 Functioning (see 3.3) Major Defect - Code No. 07002.-After three consecutive lots have met the criteria of 4.2.3.1 above, the sampling plan for this test shall be in accordance with Standard MIL-STD-105, Code letter G, double sampling plan, and an AQL of 1.5 percent. The test shall be performed as specified in 4.3.1 using equipment in accordance with 4.2.4.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and in the procedures in 4.3 (see 6.3).

4.3 Test methods and procedures.

4.3.1 Functioning.-Mines with fuzes shall be placed on firm ground so that the fuze is flush with the top of the ground. A functioning load of 500 plus 50 pounds shall be applied to the mine by any suitable means and observation made to determine compliance with 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The mines shall be packed as specified on dwg. 8830877.

5.2 Marking.-Marking of the wooden box shall be in accordance with dwg. 8830877.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

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6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
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